DDL, DCL, DML, TCL Commands

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Create a database & use :

```
1   CREATE DATABASE IF NOT EXISTS bookflow2_db;
2   USE bookflow_db;
```

Output

Action Output

#	Time	Action
1	09:08:46	SHOW DATABASES
2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
3	09:09:40	USE bookflow_db

Create a table

```
3   CREATE TABLE bank_transactions (
4       txn_id INT PRIMARY KEY,
5       customer_name VARCHAR(50),
6       branch_name VARCHAR(50),
7       transaction_type VARCHAR(20),
8       amount DECIMAL(10,2),
9       transaction_date DATE
10  );
11
```

Output

Action Output

#	Time	Action	Message
1	09:08:46	SHOW DATABASES	4 row(s) returned
2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db	1 row(s) affected
3	09:09:40	USE bookflow_db	0 row(s) affected
4	09:12:49	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, cu...	0 row(s) affected

Alter table add column

```
11  ALTER TABLE bank_transactions
12  ADD account_no VARCHAR(20);
```

Output

Action Output

#	Time	Action
1	09:08:46	SHOW DATABASES
2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
3	09:09:40	USE bookflow_db
4	09:12:49	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, cu...
5	09:13:57	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)

Alter table modify column

```
13 •  ALTER TABLE bank_transactions
14    MODIFY customer_name VARCHAR(100);
```

Output

Action Output

#	Time	Action
1	09:08:46	SHOW DATABASES
2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
3	09:09:40	USE bookflow_db
4	09:12:49	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, cu...
5	09:13:57	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)
6	09:15:57	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(1...

Rename table

```
17 •  SELECT * FROM bank_transactions;
18 •  RENAME TABLE  bank_transactions
19    TO customer_transactions;
20
```

Output

Action Output

#	Time	Action
9	09:18:07	SHOW TABLES
10	09:20:03	RENAME TABLE bank_transactions TO customer_transactions

Truncate table:

```
20 •  TRUNCATE TABLE customer_transactions;
21
```

Output

Action Output

#	Time	Action
11	09:22:08	TRUNCATE TABLE bank_transactions
12	09:22:27	TRUNCATE TABLE customer_transactions

Drop table:

```
21 •  DROP TABLE IF EXISTS bank_backup;
22
```

Output

Action Output

#	Time	Action
12	09:22:27	TRUNCATE TABLE customer_transactions
13	09:23:21	DROP TABLE IF EXISTS bank_backup

DML (DATA MANIPULATION LANGUAGE)

Create table:

```sql
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

Output

Action Output

#	Time	Action
13	09:23:21	DROP TABLE IF EXISTS bank_backup
14	09:24:21	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY,

Insert sample data:

```sql
INSERT INTO bank_transactions VALUES
(101,'Likhitha','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'Tejaswini','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'Teju','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Krishna','Vizag','Deposit',8000,'2024-01-10'),
(105,'Shiva','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Navya','Vizag','Deposit',15000,'2024-01-12'),
(107,'Kavya','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Akshitha','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Lakshmi','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Saraswati','Vijayawada','Deposit',11000,'2024-01-16');
SELECT * FROM bank_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Likhitha	Hyderabad	Deposit	5000.00	2024-01-05
102	Tejaswini	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Teju	Vijayawada	Deposit	12000.00	2024-01-08
104	Krishna	Vizag	Deposit	8000.00	2024-01-10
105	Shiva	Hyderabad	Withdrawal	3500.00	2024-01-11
106	Navya	Vizag	Deposit	15000.00	2024-01-12
107	Kavya	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Akshitha	Hyderabad	Deposit	9000.00	2024-01-14
109	Lakshmi	Vizag	Withdrawal	4000.00	2024-01-15
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Insert

```
45      (111,'Pranavi','Vizag','Deposit',7000,'2024-01-18');
46 •    SELECT * FROM bank_transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
108	Akshitha	Hyderabad	Deposit	9000.00	2024-01-14
109	Lakshmi	Vizag	Withdrawal	4000.00	2024-01-15
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
111	Pranavi	Vizag	Deposit	7000.00	2024-01-18
NULL	NULL	NULL	NULL	NULL	NULL

Update:

```
47 •    UPDATE bank_transactions
48      SET amount = 3000
49      WHERE txn_id = 105;
50 •    SELECT * FROM bank_transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_dat
104	Krishna	Vizag	Deposit	8000.00	2024-01-10
105	Shiva	Hyderabad	Withdrawal	3000.00	2024-01-11
106	Navya	Vizag	Deposit	15000.00	2024-01-12
107	Kavya	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Akshitha	Hyderabad	Deposit	9000.00	2024-01-14

Delete and select:

```
51 •    DELETE FROM bank_transactions
52      WHERE txn_id = 111;
53 •    SELECT * FROM bank_transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
107	Kavya	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Akshitha	Hyderabad	Deposit	9000.00	2024-01-14
109	Lakshmi	Vizag	Withdrawal	4000.00	2024-01-15
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Select with where:

```
54 •    SELECT *
55      FROM bank_transactions
56      WHERE transaction_type='Deposit';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_d
101	Likhitha	Hyderabad	Deposit	5000.00	2024-01-05
103	Teju	Vijayawada	Deposit	12000.00	2024-01-08
104	Krishna	Vizag	Deposit	8000.00	2024-01-10
106	Navya	Vizag	Deposit	15000.00	2024-01-12
108	Akshitha	Hyderabad	Deposit	9000.00	2024-01-14
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Select with order by:

```
57 •  SELECT *
58    FROM bank_transactions
59    ORDER BY amount DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Navya	Vizag	Deposit	15000.00	2024-01-12
103	Teju	Vijayawada	Deposit	12000.00	2024-01-08
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
108	Akshitha	Hyderabad	Deposit	9000.00	2024-01-14
104	Krishna	Vizag	Deposit	8000.00	2024-01-10
101	Likhitha	Hyderabad	Deposit	5000.00	2024-01-05
109	Lakshmi	Vizag	Withdrawal	4000.00	2024-01-15
105	Shiva	Hyderabad	Withdrawal	3000.00	2024-01-11
102	Tejaswini	Hyderabad	Withdrawal	2000.00	2024-01-06
107	Kavya	Vijayawada	Withdrawal	1000.00	2024-01-13
NULL	NULL	NULL	NULL	NULL	NULL

DCL (DATA CONTROL LANGUAGE)

GRANT ALL PRIVILEGES:

```
60 •  CREATE USER 'BranchManager'@'localhost'
61    IDENTIFIED BY 'Branch@123';
62 •  GRANT ALL PRIVILEGES
63    ON bookflow1_db.bank_transactions
64    TO 'BranchManager'@'localhost';
65 •  SHOW GRANTS FOR 'BranchManager'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

Grants for BranchManager@localhost
GRANT USAGE ON *.* TO `BranchManager`@`...
GRANT ALL PRIVILEGES ON `bookflow1_db`.`b...

Revoke all:

```
66 •  REVOKE ALL PRIVILEGES
67    ON bookflow1_db.bank_transactions
68    FROM 'BranchManager'@'localhost';
```

Output

Action Output

#	Time	Action
34	09:37:33	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'Branch...
35	09:37:35	GRANT ALL PRIVILEGES ON bookflow1_db.bank_transactions TO 'Br...
36	09:37:37	SHOW GRANTS FOR 'BranchManager'@'localhost'
37	09:38:45	REVOKE ALL PRIVILEGES ON bookflow1_db.bank_transactions FRO...

TCL (TRANSACTION CONTROL LANGUAGES)

Commit:

```
69 •  UPDATE bank_transactions
70    SET amount=6000
71    WHERE txn_id=101;
72 •  COMMIT;
```

Output

Action Output

#	Time	Action
36	09:37:37	SHOW GRANTS FOR 'BranchManager'@'localhost'
37	09:38:45	REVOKE ALL PRIVILEGES ON bookflow1_db.bank_transactions FRO...
38	09:39:50	UPDATE bank_transactions SET amount=6000 WHERE txn_id=101
39	09:39:52	COMMIT

Savepoint:

```
73 •  SAVEPOINT Before_Update;
```

Output

Action Output

#	Time	Action
37	09:38:45	REVOKE ALL PRIVILEGES ON bookflow1_db.bank_transactions FRO.
38	09:39:50	UPDATE bank_transactions SET amount=6000 WHERE txn_id=101
39	09:39:52	COMMIT
40	09:40:55	SAVEPOINT Before_Update

Rollback to savepoint:

```
75    SET amount=99999
76    WHERE txn_id=102;
77 •  SELECT * FROM bank_transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Likhitha	Hyderabad	Deposit	6000.00	2024-01-05
102	Tejaswini	Hyderabad	Withdrawal	99999.00	2024-01-06
103	Teju	Vijayawada	Deposit	12000.00	2024-01-08
104	Krishna	Vizag	Deposit	8000.00	2024-01-10
105	Shiva	Hyderabad	Withdrawal	3000.00	2024-01-11
106	Navya	Vizag	Deposit	15000.00	2024-01-12
107	Kavya	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Akshitha	Hyderabad	Deposit	9000.00	2024-01-14
109	Lakshmi	Vizag	Withdrawal	4000.00	2024-01-15
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL